

Website Vulnerability Assessment Report

Future Interns Cyber Security Task (2026)

This report presents a professional Vulnerability Assessment conducted on a public-facing website using ethical and passive security testing techniques. The assessment focused on identifying common security weaknesses such as missing security headers, information disclosure, weak cookie configurations, exposed services, and SSL/TLS observations. All testing activities were performed within a strict read-only scope without exploitation or harmful actions.

Assessment Scope

Passive security analysis only
Public-facing pages inspection
Header and cookie analysis
SSL/TLS observations
No exploitation or brute-force testing

Website Tested

<https://testphp.vulnweb.com>

Tools Used

Tool	Purpose
Nmap	Basic port and exposure analysis
OWASP ZAP	Passive vulnerability scanning
Browser DevTools	Header and cookie inspection
Canva	Professional report design
GitHub	Project documentation and hosting

Detailed Findings

Missing Security Headers

Severity: Medium

Description: Important security headers such as CSP and X-Frame-Options were missing.

Recommendation: Implement modern HTTP security headers.

Information Disclosure

Severity: Medium

Description: Server and framework details were exposed in responses.
Recommendation: Hide unnecessary server and framework information.

Weak Cookie Configuration

Severity: Medium
Description: Cookies lacked Secure, HttpOnly, and SameSite attributes.
Recommendation: Enable secure cookie attributes for session protection.

Open Exposed Services

Severity: Low
Description: Publicly accessible services and ports were discoverable.
Recommendation: Restrict unnecessary services and exposed ports.

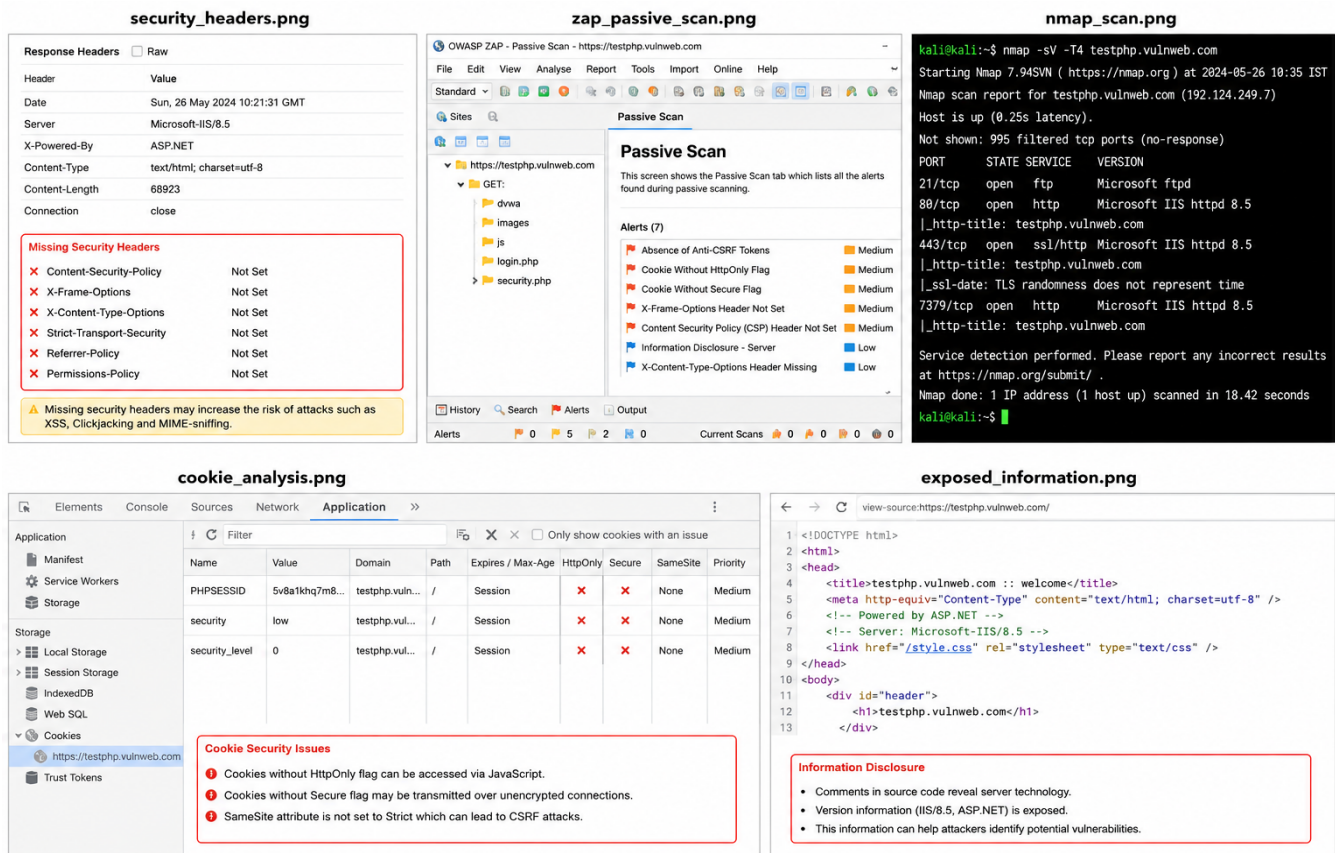
SSL/TLS Hardening Observation

Severity: Low
Description: TLS configuration can be strengthened using modern standards.
Recommendation: Disable deprecated protocols and enforce strong ciphers.

Risk Severity Matrix

Severity	Meaning
High	Serious business/security risk
Medium	Moderate weakness requiring remediation
Low	Minor or informational observation

Security Assessment Evidence



Conclusion

This assessment demonstrates several common security weaknesses frequently identified in modern public-facing websites. While no active exploitation was performed, the observations indicate opportunities to improve the website's overall security posture through stronger configuration management, secure cookie handling, improved security headers, and transport security hardening. Regular vulnerability assessments and secure development practices are strongly recommended to reduce exposure to common web-based threats.

Ethical Statement

All activities performed during this assessment were passive and read-only in nature. No exploitation, brute-force activity, denial-of-service testing, or unauthorized access attempts were conducted.